ModelIt!

Teacher's Guide

How Self-Driving Cars See the Road

Modeling the Sensor Fusion and Decision Systems of Autonomous Vehicles

Quick Reference

Grade Level	Grade 10L3
NGSS Standard	HS-PS4-2, HS-PS4-5
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-PS4-2, HS-PS4-5

Evaluate questions about the advantages of using digital transmission and storage of information. Communicate technical information about how some technological devices use the principles of wave behavior and wave interactions with matter to transmit and capture information and energy.

Three-Dimensional Learning

Science Practice	**Communicating Technical Information** — Students communicate technical information about how LIDAR, radar, and cameras use wave behavior to detect and measure the vehicle's environment.
Disciplinary Core Idea	**PS4.C Information Technologies and Instrumentation** — Students model how electromagnetic waves of different frequencies (laser, radio, visible light) are used to detect, measure, and classify objects for autonomous vehicle navigation.
Crosscutting Concept	**Systems and System Models** — Students model the autonomous driving system as interacting subsystems (sensing, processing, deciding, acting) where the performance of each subsystem depends on and affects the others.

CAST Testing Alignment

MULTIPLE CHOICE:

During heavy rain, a self-driving car's LIDAR performance drops by 60% and camera performance drops by 40%, but radar performance drops by only 10%. According to your model, how does the sensor fusion system maintain Obstacle Classification accuracy?

CONSTRUCTED RESPONSE:

Using your model, explain why a self-driving car that performs perfectly in 99.9% of situations is still not safe enough for public roads. Address the concept of edge cases, the scale of driving (millions of miles), and the difference between statistical safety and individual trust in your response.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Human drivers kill over 40,000 people per year in the United States alone, with 94% of crashes caused by human error — distraction, impairment, fatigue, and poor judgment. Autonomous vehicles promise to eliminate these errors with sensors that never blink, algorithms that never get tired, and reaction times five times faster than humans. But replacing human perception and decision-making with technology is one of the hardest engineering challenges ever attempted.

The Sensor Stack

Modern autonomous vehicles use four complementary sensor types: LIDAR (laser-based 3D mapping, centimeter accuracy, limited in rain/snow), cameras (color/texture recognition, sign reading, lane detection, limited in darkness/glare), radar (velocity measurement, all-weather capability, low resolution), and ultrasonic (short-range parking and close-object detection). Each sensor has strengths and weaknesses — LIDAR excels at precise distance measurement but cannot read signs, cameras read signs but cannot measure distance as precisely, radar works in all weather but provides only coarse object shapes.

Sensor Fusion: The Key Innovation

Sensor fusion is the algorithmic process of combining all sensor data into a unified perception model. When LIDAR detects an object at 150 meters, the camera classifies it as a pedestrian, and radar measures its approach velocity at 5 km/h. No single sensor provides all three pieces of information. Fusion algorithms use Kalman filters, neural networks, and probabilistic models to weight each sensor's contribution based on its confidence and current reliability — automatically reducing weight on rain-degraded LIDAR while increasing reliance on radar.

The Decision Pipeline

From detection to action, the autonomous driving pipeline has five stages: (1) Perception — raw sensor data converted to environmental model (50-100 ms), (2) Prediction — estimating what every detected object will do next (10-30 ms), (3) Planning — computing the optimal path avoiding all predicted hazards (20-50 ms), (4) Control — translating the planned path into steering, braking, and acceleration commands (10-20 ms), (5) Actuation — physical execution of commands by the vehicle systems (20-50 ms). Total pipeline: 110-250 ms, compared to 1,500+ ms for human perception-to-action.

The Edge Case Problem

Autonomous systems are trained on millions of miles of driving data, but the real world is infinitely variable. Edge cases — unusual scenarios not well-represented in training data — expose the fundamental limitation of pattern-matching approaches. A construction worker wearing a dinosaur costume, an overturned wheelchair in a lane, a drone flying at car height, or a child darting between parked cars all challenge classification algorithms in ways that human general intelligence handles naturally. The industry debates whether edge cases can ever be fully eliminated or whether a different approach to AI is needed.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify nine key components of the autonomous vehicle perception and decision system: LIDAR pulse frequency, detection range, signal processing speed, obstacle classification, weather interference, decision latency, sensor fusion accuracy, route optimization, and safety margin.

E **Establish Relationships**

Students map relationships showing that LIDAR pulse frequency and weather interference determine detection quality, which feeds into obstacle classification and sensor fusion accuracy, which determine decision latency and safety margin — with weather creating a degradation cascade through multiple pathways.

V **Visualize & Model**

Students build a computational model showing the nine-component autonomous driving system with sensor redundancy and weather degradation pathways.

E **Evaluate Outcomes**

Students run clear highway, heavy rain urban, and edge case scenarios to test their model and identify the conditions where autonomous driving transitions from safe to unreliable.

R **Revise & Extend**

Students add V2X communication, map precision, or cybersecurity resilience to model more complete autonomous vehicle system dynamics.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Your eyes have 120 million photoreceptors. This car has 64 laser beams, 8 cameras, 6 radars, and 12 ultrasonics — processing 1 terabyte of data per hour. But can it see better than you?"*

Watch For

Show a split-screen: human driver's view versus the car's sensor visualization (point clouds, bounding boxes, trajectory predictions). The complexity hooks students.

Slide 2: Learning Objectives

3 min

Say: *"Today you are engineering the perception system that must be better than human vision in every condition — rain, darkness, fog, and the completely unexpected."*

Watch For

Have students read objectives. Pre-teach 'sensor fusion' and 'LIDAR.' Demonstrate LIDAR concept with a flashlight in a dark room — light hits objects and bounces back, revealing their distance.

Slide 3: Big Question

3 min

Say: *"40,000 people die on US roads every year. 94 percent from human error. Can machines do better? And what happens when they cannot?"*

Watch For

Quick-write: Students list situations where a self-driving car would outperform a human driver, and situations where it would fail. Share out and create two lists.

Slide 4: LEVER Framework

2 min

Say: *"We are building a model of a machine that must make life-or-death decisions in 200 milliseconds based on noisy, incomplete sensor data. No pressure."*

Watch For

Preview LEVER steps. Emphasize the sensor fusion concept — no single sensor is sufficient, but together they create something more capable than any one.

Slide 5: Activity 1: Components

8 min

Say: *"Nine variables. Two you control through engineering design. Seven that the physics and software determine. The challenge is making them work together faster than any human can think."*

Watch For

Guide students through all nine components. Spend extra time comparing what each sensor type provides — LIDAR for distance, camera for classification, radar for velocity and weather resilience.

Slide 6: Activity 2: Connections

8 min

Say: *"When it rains, LIDAR degrades. When LIDAR degrades, the 3D map gets fuzzy. When the map gets fuzzy, obstacle classification confidence drops. Trace the cascade."*

Watch For

Help students map the weather degradation cascade through multiple sensors. Then show how sensor fusion mitigates by reweighting sensor trust. Use green arrows for fusion recovery pathways.

Slide 7: Activity 3: Simulation

10 min

Say: *"Perfect conditions, terrible weather, and the thing no algorithm was trained for. Let us see where the system works, where it struggles, and where it fails."*

Watch For

Students predict system performance for each scenario. Run simulations. The edge case scenario should generate discussion about the limits of pattern-matching AI.

Slide 8: Discoveries

5 min

Say: *"Self-driving cars are superhuman in routine driving and subhuman in the unexpected. The question is whether we can make the superhuman common enough and the failures rare enough to save more lives than we lose."*

Watch For

Lead discussion on the statistical safety argument: if autonomous vehicles are 10 times safer overall but fail catastrophically in rare edge cases, is that acceptable? Who decides?

Slide 9: STEM Challenge

12 min

Say: *"Your car company wants to launch in Seattle — rain capital of the US. Design a sensor system that works when every sensor is degraded. Or tell the car to stop."*

Watch For

Groups design weather-resilient sensor fusion architectures. Include handoff-to-human protocols. Present with model evidence for performance claims. Class evaluates which design is safest.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** LIDAR Pulse Frequency, Weather Interference
- **INTERNAL:** Detection Range, Signal Processing Speed, Obstacle Classification, Decision Latency, Sensor Fusion Accuracy, Route Optimization, Safety Margin

Relationships

- **LIDAR Pulse Frequency to Detection Range:** POSITIVE (+) — Higher pulse frequency creates denser point clouds that can resolve smaller and more distant objects, effectively extending the reliable detection range. At 300,000+ pulses per second, modern LIDAR can detect and classify objects at 200+ meters.
- **Weather Interference to Sensor Fusion Accuracy:** NEGATIVE (-) — Increased weather interference degrades individual sensor performance — LIDAR scatters in rain, cameras lose contrast in fog, radar gets surface reflections in snow. When multiple sensors degrade simultaneously, the fusion algorithm has fewer reliable data sources, reducing overall perception accuracy.
- **Sensor Fusion Accuracy to Decision Latency:** NEGATIVE (-) — Higher sensor fusion accuracy means the decision system receives clear, confident object classifications, enabling faster path planning. When accuracy drops due to weather or edge cases, the system requires additional processing cycles to resolve ambiguity, increasing total decision latency.

Key Discoveries

- Sensor fusion provides redundancy that no single sensor can match — when LIDAR fails in rain, radar takes over for distance measurement while cameras maintain object classification
- Decision Latency for autonomous systems (100-300 ms) is 5-10 times faster than human reaction time (1,500+ ms), but this advantage narrows significantly in complex, ambiguous situations requiring judgment
- Edge cases represent the fundamental challenge: autonomous systems excel at common scenarios but struggle with rare, novel situations that humans handle through general intelligence and common sense
- The Safety Margin variable reveals a deep engineering trade-off: larger margins make the car safer but also slower, less efficient, and more disruptive to traffic flow — finding the optimal margin is both an engineering and ethical question

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Self-driving cars use cameras like human eyes"	While cameras provide color and texture information similar to human vision, they are only one of four sensor types. LIDAR provides 3D spatial information far more precise than human depth perception. Radar measures object velocities directly. The autonomous system perceives in 360 degrees simultaneously, in wavelengths humans cannot see, and processes data from all sensors in parallel. The result is fundamentally different from human vision — more precise in some ways (exact distances, 360-degree coverage, no blind spots) and less capable in others (general object understanding, context, common sense).
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Autonomous cars are safer because computers are faster"	Raw speed is necessary but not sufficient. A computer can react in 150 ms versus a human's 1,500 ms, but safety also requires accurate perception, correct classification, and appropriate decisions. A computer that reacts in 150 ms to a misclassified object (slamming the brakes for a plastic bag it thinks is a boulder) is not safer than a human who takes 1,500 ms but correctly ignores the harmless bag. Speed without accuracy can be more dangerous than slower, correct human judgment.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Self-driving cars will eliminate all traffic accidents"	Even perfect autonomous vehicles cannot eliminate all accidents. Some crashes involve unavoidable mechanical failures (tire blowouts), road infrastructure failures (bridge collapses), or extreme weather events. Additionally, during the transition period when autonomous and human-driven vehicles share roads, the interaction between predictable machine behavior and unpredictable human behavior creates new failure modes. Most safety researchers predict autonomous vehicles will reduce crashes by 80-90%, not eliminate them entirely.